

Musculoskeletal System

Cont...

Objectives

- After completion of this module you will be able to:
 - ✓ Describe the structure and location of bones in the body
 - ✓ Describe the structure and location of muscles in the body
 - ✓ Define terms related to bone structure, joints, and joint movements
 - ✓ Define terms related to structure of muscles

Cont....

- The musculoskeletal system is made up of:
 - Bones (the skeleton)
 - Cartilage
 - Tendons
 - ligaments,
 - Joints
 - Muscles
 - other connective tissues



Function

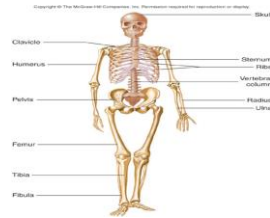
- Give **shape** and **structure** to the body
- provide support
- Allow movement
- Protect internal organs
- Store calcium and other minerals
- Produce certain blood cells
- Produce heat
- subdivided into two broad systems:
 - Skeletal system
 - Muscular system

Cont....



Skeletal system

- is all the **bony** tissues in the body
- ✓ **Components** of the skeletal system
 - Bones
 - Joints
 - Cartilages
 - **Ligaments**
 - **tendon**



- The skeleton has ~206 bones

Axial skeleton

- Skull
 - Cranium = 8
 - Face = 14
- Hyoid = 1
- Auditory ossicles = 6
- Vertebral column = 26
- Thorax
 - Sternum = 1
 - Ribs = 24
- Subtotal = 80**

Cont....



Appendicular skeleton

- Pectoral (shoulder) girdles
 - Clavicle = 2
 - Scapula = 2
- Upper limbs
 - Humerus = 2
 - Ulna = 2
 - Radius = 2
 - Carpals = 16
 - Metacarpals = 10
 - phalanges = 28

Cont....



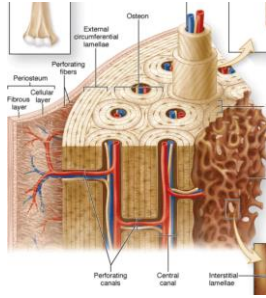
- Pelvic (hip) girdle
 - Hip(pelvic bone) = 2
- Lower limbs
 - Femur = 2
 - Patella = 2
 - Fibula = 2
 - Tibia = 2
 - Tarsals = 14
 - Metatarsals = 10
 - Phalanges = 28
 - subtotal = 126
- Total in the **adult** skeleton = Axial skeleton **80** + Appendicular skeleton **126** = 206

Cont....



Microscopic Anatomy of Bone

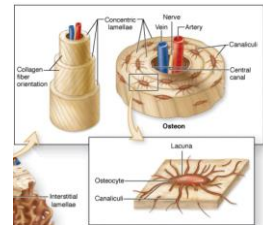
- **Osteon (Haversian System)**
 - A unit of bone
 - **Central (Haversian) canal**
 - Opening in the **center** of an osteon
 - Carries blood **vessels** and **nerves**
- **Perforating (Volkman's) canal**
- Canal **perpendicular** to the central canal
- Carries blood vessels and nerves



Cont....

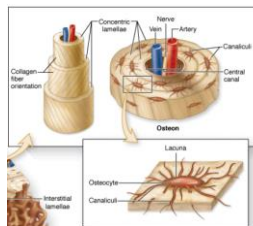
Bone Cells

- **Osteocytes**
 - **Mature** bone cells
 - Contained within the spaces in the bone matrix called **lacunae**
 - extends numerous processes into small tunnels called **canaliculi**



Cont....

- **Osteoclasts**
 - **bone-resorbing** cells present on bone surfaces
 - **Bone-destroying** cells
 - Break down bone matrix for **remodeling** and **release of calcium**.
- ✓ Bone remodeling is a process by both **osteoblasts** and **osteoclasts**



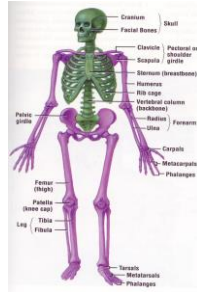
Classification of bones

According to their Location

- Divided into two:
 - Axial Skeleton
 - Appendicular Skeleton

I. Appendicular Skeleton

- is made up of the bones of the **limbs** and their **girdles**
- ✓ **Pectoral girdles** attach the upper limbs to the body trunk
- ✓ **Pelvic girdle** secures the lower limbs



Pectoral Girdles

- Shoulder Girdles
- consist of the **clavicles** and the **scapulae**
- attach the **upper** limbs to the **axial** skeleton
- provide **attachment** points for **muscles** that move the upper limbs
- articulates with the **axial** skeleton only at the **sternoclavicular** joint, allowing great mobility.
- supported, stabilized, and moved by **axio appendicular** muscles



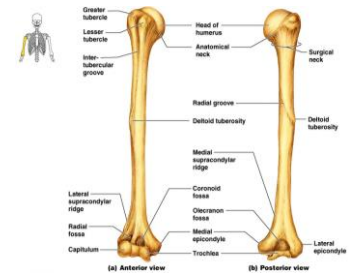
Bones of the upper limb

- Each upper limb consists of 30 bones
- humerus = in the arm
- ulna and radius = in the forearm
- 8 carpals = wrist bones
- 5 metacarpals = palm bones
- 14 phalanges = finger bones in the hand



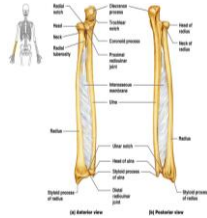
Humerus

- Is the **largest** bone in the **upper** limb
- articulates with the **scapula**,
- at **glenohumeral** joint
- the **radius** and **ulna** at the **elbow** joint.



Bones of Forearm

- are the **radius** and **ulna**
- They articulate **proximally** with the **humerus** and **distally** with the **wrist bones**
- They also articulate with each other proximally and distally at small **radioulnar** joints
- **Interosseous membrane** connects the two bones along their entire length



Bones of Hand

- **Skeleton of the hand** contains **wrist bones** (8 carpals), **bones of the palm** (metacarpals), and **bones of the fingers** (phalanges)



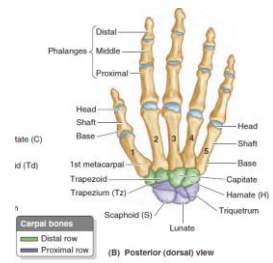
Cont...

- **Carpal** bones are arranged in **proximal** and **distal** rows of four.
- From **lateral** to **medial**, the four carpal bones in the **proximal** row are:
- Scaphoid
- Lunate
- Triquetrum
- Pisiform



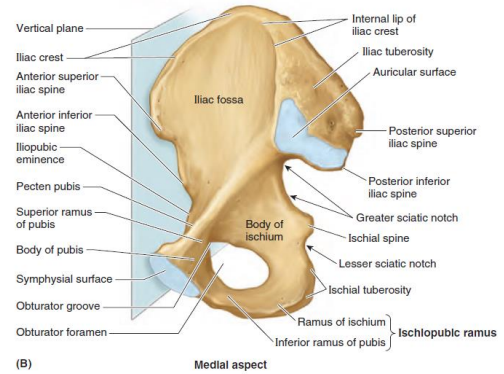
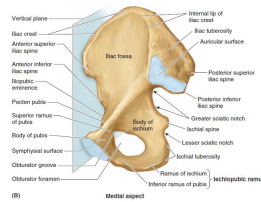
Cont...

- From **lateral** to **medial**, the four carpal bones in the **distal** row are the:
- Trapezium
- Trapezoid
- Capitate
- Hamate



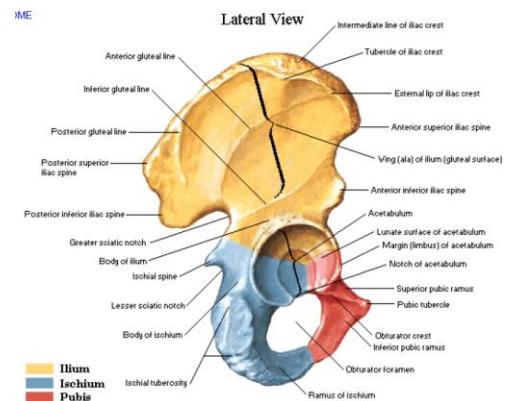
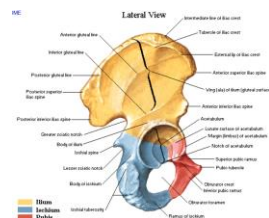
Pelvic (hip) girdle

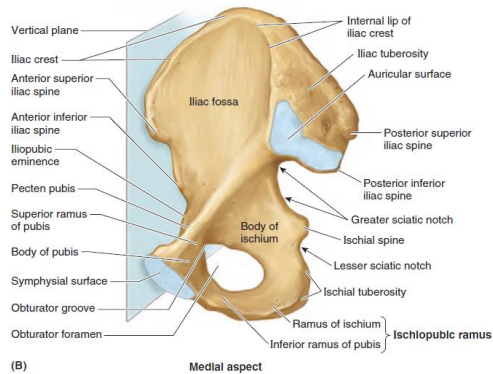
- consists of the two **hip** bones, also called coxal bones
- hip bones of a newborn is composed of three parts: the **ilium**, the **pubis**, and the **ischium**
- Mature hip bone is formed by the **fusion** of these three primary bones at the end of the **teenage** years



Ilium

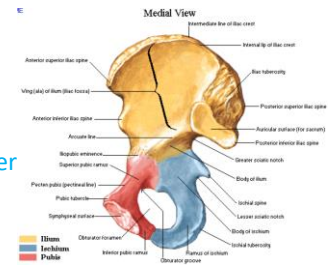
- forms the **largest** part of the hip bone
- **Structural features:**
 - **ala**, **body** of the ilium, **anterior superior** and **anterior inferior iliac spines**, **iliac crest**, **gluteal lines**, **iliac fossa**





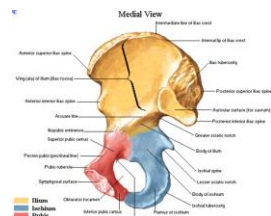
- forms the **postero-inferior** part of the hip bone
- Its ramus joins the inferior ramus of the pubis to form the **ischiopubic ramus**.
- Posteriorly forms the **inferior margin greater sciatic notch**.
- **Structural features:** ischial spine, lesser sciatic notch, ischial tuberosity.

Ischium



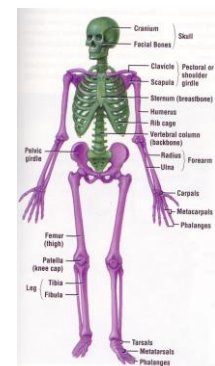
Pubis

- forms the **anteromedial** part of the hip bone.
- **Structural features:** -body, superior and inferior rami, pubic tubercles, pecten pubis.



Bones of the lower limb

- Each lower limb is composed of 30 bones:
 - femur = in the thigh
 - patella = kneecap
 - tibia and fibula = in the leg
 - 7 tarsals = ankle bones
 - 5 metatarsals
 - 14 phalanges = toes in the foot



Thigh bone

- Femur
- is the **longest** and **heaviest** bone in the body.
- It **transmits** body **weight** from the hip bone to the **tibia** when a person is standing.
- consists of a **shaft** , **proximal** and **distal** ends.
- The **proximal** end of the femur consists of a **head**, **neck**, and two **trochanters** - **greater** and **lesser**.



Bones of leg

- **Tibia and Fibula**
- The **tibia** articulates with the **condyles** of the **femur** superiorly and the **talus** inferiorly.
- The **fibula** mainly functions as an **attachment** for **muscles**, - but it is also important for the **stability** of the **ankle joint**.
- The **shafts** of the tibia and fibula are connected by a dense **interosseous membrane**



Bones of Foot

- include the **tarsus**, **metatarsus**, and **phalanges**.
- There are **7 tarsal** bones, **5 metatarsal** bones, and **14 phalanges**.
- The **tarsus** consists of: **talus**, **calcaneus**, **cuboid**, **navicular**, and three **cuneiforms**.

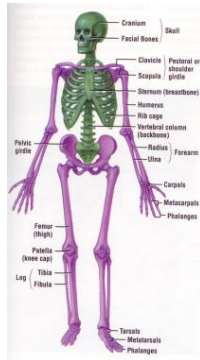


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- The **metatarsus** consists of five **metatarsals** that are numbered from the **medial** side.
- There are **14 phalanges**.

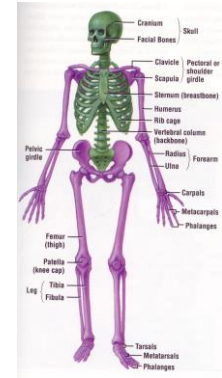


Axial Skeleton



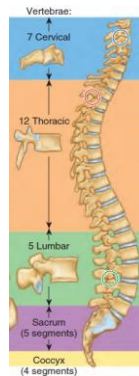
Axial Skeleton

- Forms the **longitudinal** part of the body
- Divided into **three** parts
 - Skull
 - Vertebral column
 - Bony thorax

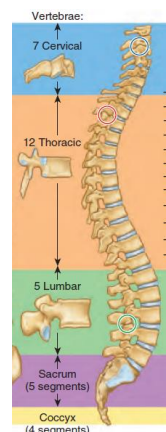


Vertebral Column

- Consists of **33** vertebrae arranged in **five** **regions**:
 - 7 cervical
 - 12 thoracic
 - 5 lumbar
 - 5 sacral
 - 4 coccygeal
- sacral vertebrae fuse to form **sacrum**
- coccygeal vertebrae fuse to form **coccyx**



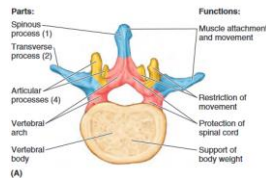
- The vertebrae in the vertebral column are separated by resilient **IV discs**
- Significant **motion** occurs only b/n the **25** superior vertebrae
- Each vertebrae is given a name according to its **location**



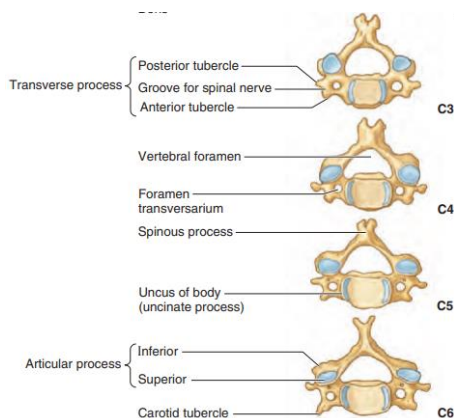
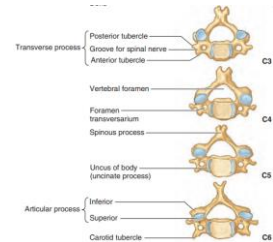
Cont....

basic structure of vertebrae

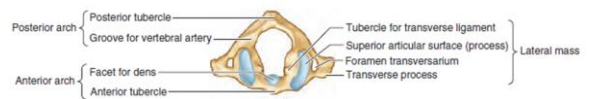
- A **typical** vertebra consists of:
 - vertebral body
 - a vertebral arch
 - seven processes

**Cervical Vertebrae**

- skeleton of the **neck**
- Located b/n the **cranium** and **thoracic** vertebrae
- oval **transverse foramen** is their distinctive feature
- Vertebrae **C3–C7** are **typical** cervical vertebrae
- vertebrae **C1** and **C2** are **atypical**

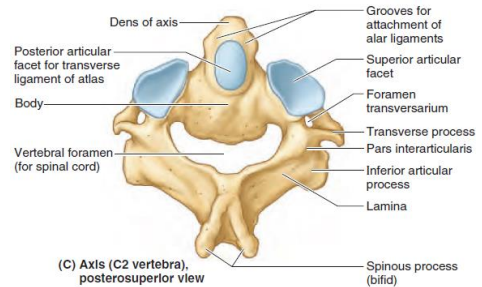
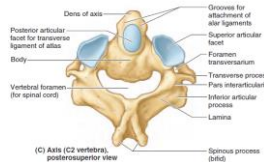
**Vertebra C1 (atlas)**

- has neither a **body** nor a **spinous process**
- articulations b/n the superior articular surfaces of the lateral masses of the **atlas** and the **occipital condyles** known as **Atlanto-occipital Joints**
- These joints permit nodding of the head indicating the **"yes"** movement



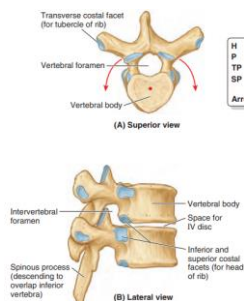
Vertebra C2(axis)

- Is strongest
- The **odontoid process** is it's distinguishing feature
- **Atlanto-axial** Joints b/n C1 and C2 permits the head to be turned from side to side to indicate the **"no"** movement



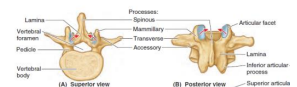
Thoracic Vertebrae

- 12 bones in the region of thorax
- increase in size from T1 to T12
- serve as attachment for the ribs
- have extra articular process for attachment of ribs
- have long spinous processes that slope obliquely downward



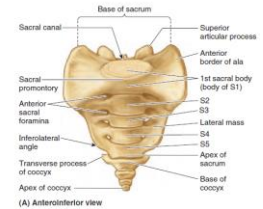
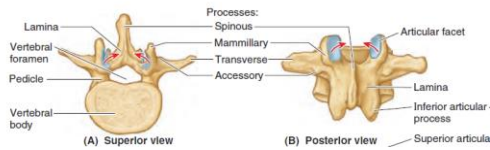
Lumbar vertebrae

- Are b/n the **thorax** and **sacrum**.
- are the **largest** vertebrae of the column.
- have large **bodies**, and **thick** processes



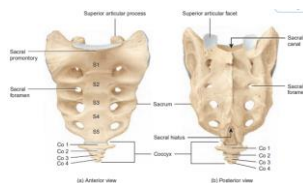
Sacral vertebrae

- 5 fused vertebrae
- is located between the **hip** bones
- provides **sacroiliac joint**



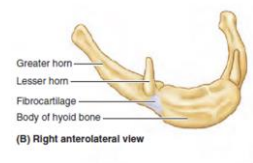
Coccyx

- called the **tail** bone
- composed of **4 or 5** fused vertebrae



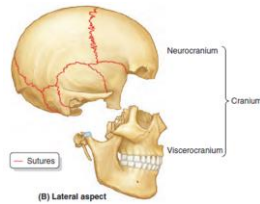
Hyoid bone

- Lies in the **anterior** part of the neck at the level of the **C3** vertebra.
- U-shaped bone suspended by muscles.
- Does **not** articulate with any other bone



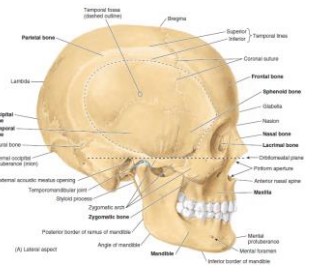
Skull (cranium)

- is the **skeleton** of the head
- Has two parts:
 1. **Neurocranium**
- bony case of:
 - Brain
 - cranial meninges
 - proximal parts of the cranial nerves
 - the vasculature of the brain

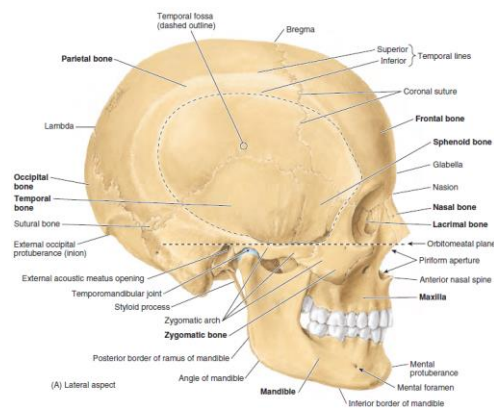


- in adults is formed by a series of **eight** bones

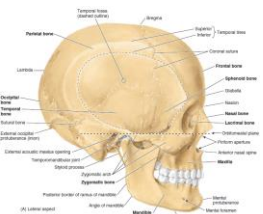
- Frontal
 - Ethmoidal
 - Sphenoidal
 - Occipital
- singular bones
- Temporal
 - Parietal
- bilateral pairs



Cont....



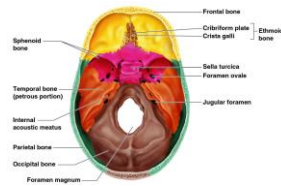
- **Frontal bone** : forms anterior roof of cranium, nasal cavity & orbit
- **Parietal bone**: separated from frontal bone by coronal suture
- **Occipital bone**: forms the back & much of the base of the skull
- **Temporal bone**: forms lower sides of the cranium



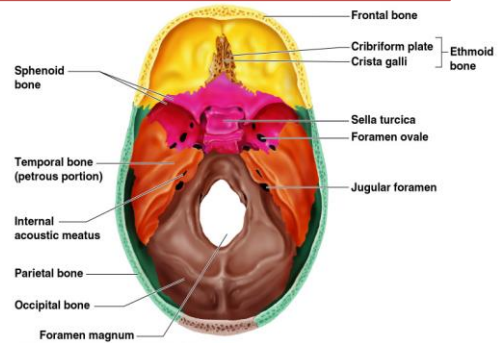
Cont....

Cont....

- **Sphenoid bone**: keystone bone (all other bones articulate with it)
- shaped like bat with outstretched wings
- completes sides of skull and contributes to orbits
- **Ethmoid bone**: lies in front of sphenoid
- forms orbits and nasal septum



Human Skull, Superior View



Cont....

2. Viscerocranium (facial skeleton)

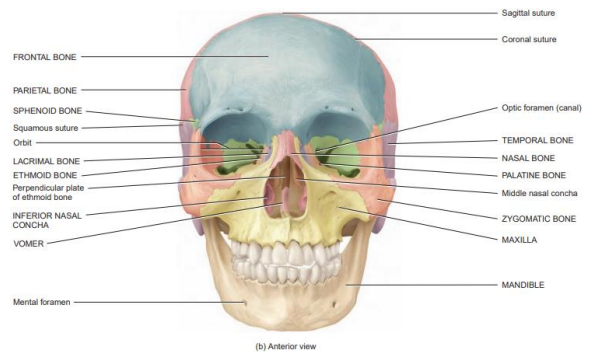
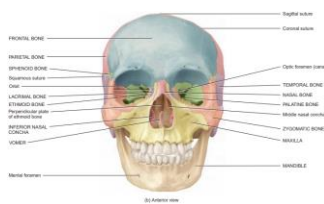
- consists of 15 irregular bones

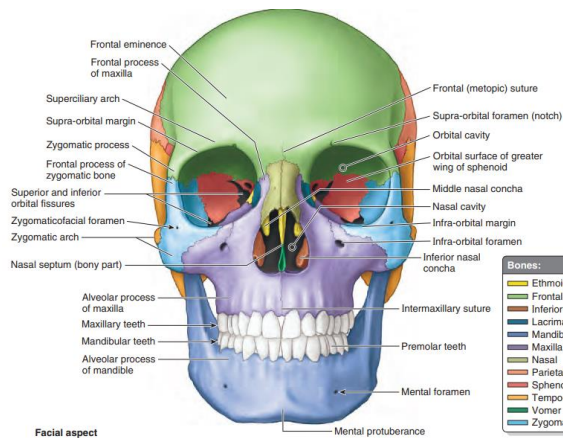
- Mandible
- Ethmoid
- Vomer bones

Singular

- ✓ Maxillae
- ✓ inferior nasal conchae
- ✓ Zygomatic
- ✓ Palatine
- ✓ nasal
- ✓ lacrimal bones

bilateral pairs





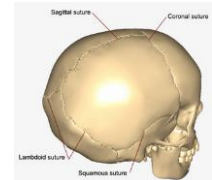
Unique Features of the Skull

A. Sutures

- immovable joints in most cases in an adult
- holds skull bones together

I. Coronal suture

- unites the **frontal** bone and two **parietal** bones



II. Sagittal suture

- unites the **two parietal** bones

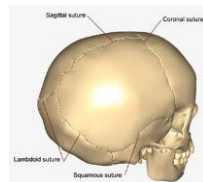
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III. Lambdoid suture

- unites the **parietal** bones to the **occipital** bone

IV. Squamous sutures :

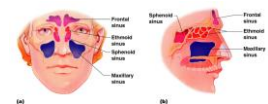
- unite the **parietal** bones to the **temporal** bone



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B. Paranasal sinuses

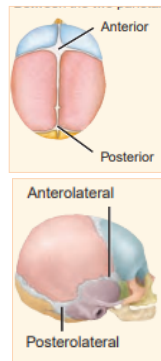
- Paired cavities, near the nasal cavity
- Include **frontal** sinus, **sphenoid** sinus, **ethmoidal** sinuses, and **maxillary** sinuses
- produce **mucus**
- serve as **resonating** chambers
- **lighten** the weight of the skull



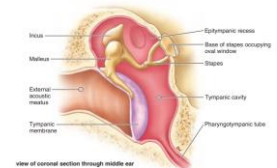
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C. Fontanelles

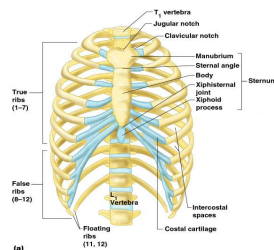
- found b/n cranial bones at birth
- include **anterior** fontanel, **posterior** fontanel, **anterolateral** fontanel, and **posterolateral** fontanel
- enable the fetal skull to be **compressed** as it passes through the birth canal
- permit rapid **growth** of the brain during infancy

**Auditory Ossicles**

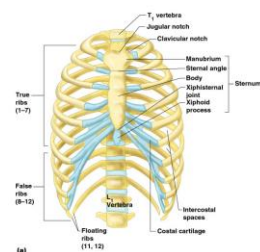
- form a mobile chain of small bones across the tympanic cavity from the **tympanic membrane** to the **oval window**
- Three auditory ossicles in the middle ear of each ear
 - **malleus**
 - **incus**
 - **stapes**
- Helps to transmit sound impulses

**Bony Thorax**

- Forms a cage to protect major organs
- Made-up of three parts
 - **Sternum**
 - **Ribs**
 - **Thoracic vertebrae**

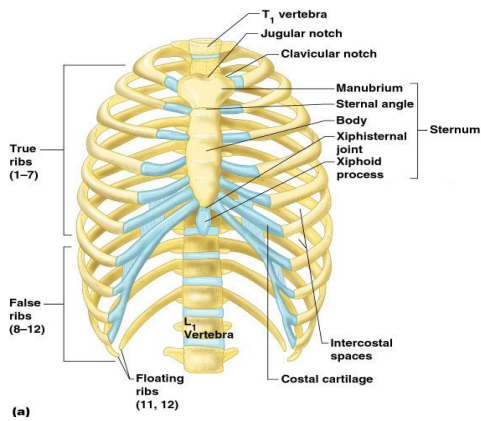
**Ribs**

- 12 pairs (connect directly to **thoracic vertebrae** in the back)
- 7 pairs "**true ribs**" connect directly to sternum anteriorly
- the remaining 5 pairs are called "**false ribs**"
- last 2 pairs "**floating ribs**" do not attach to sternum at all



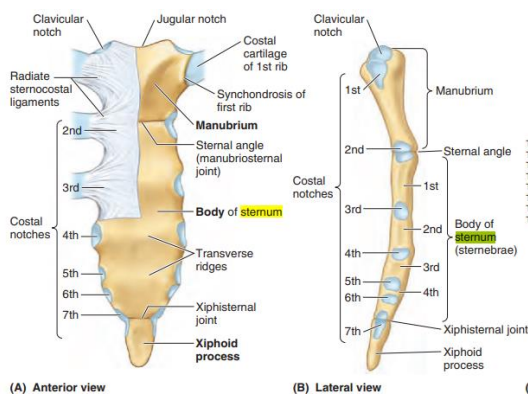
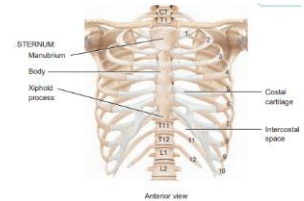
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Sternum

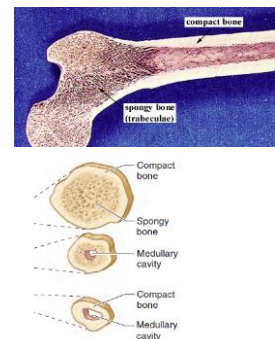
- Flat bone located anteriorly
- Has 3 fused bones: **manubrium, body & xiphoid process**
- On its side, it has **costal & clavicular notches** and superiorly **jugular notch**
- **Sternal angle** is palpated between **manubrium & body**



II. According to their Structural Appearances

- Compact (Dense) Bone
- Spongy(trabecular, cancellous) Bone

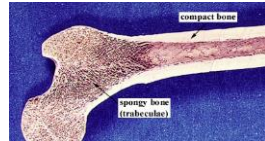
- distinguished by the relative amount of **solid matter**
- by the **number** and **size** of the **spaces** they contain



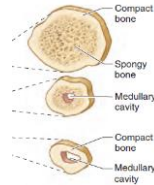
Cont....

➤ **Compact bone**

- Have Outer dense layer that looks smooth and homogenous to the naked eye.
- Have **medullary** cavity.

➤ **Spongy Bone**

- **Inner** layer composed of small needle like **spicules** of bone and numerous **opening** having a **spongy** appearance to the naked eye.



III. According to their Shapes

➤ **Long Bones**

- Typically **longer** than **wide**
- Have a **shaft** with **heads** at both ends
- Contain **mostly compact** bone

- are **tubular**

Examples: Femur, humerus

➤ **Short Bones**

- Generally **cube-shape**
- Contain mostly **spongy** bone

Examples: Carpals, tarsals



Cont....

➤ **Flat Bones**

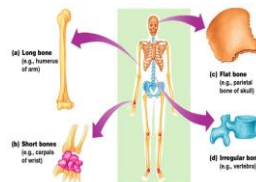
- **Thin** and **flattened**
- Usually **curved**
- Thin layers of **compact** bone around a layer of **spongy** bone.

Examples: Skull, ribs, sternum

➤ **Irregular Bones**

- **Irregular** shape
- have various shapes **other** than **long, short, or flat**.
- Do **not** fit into **other bone** classification categories.

Example: Vertebrae and hip



Cont....

➤ **Sesamoid bones**

- Develop in certain **tendons**
- found where **tendons** **cross** the ends of **long bones** in the limbs.
- they protect the tendons from **excessive wear**

Examples: the patella or knee cap

Gross Anatomy of a Long Bone

• Diaphysis

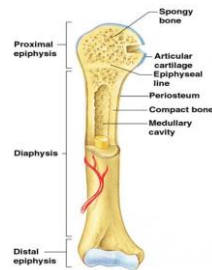
- Shaft
- Composed of compact bone

• Epiphysis

- Ends of the bone
- Composed mostly of spongy bone

• Articular cartilage

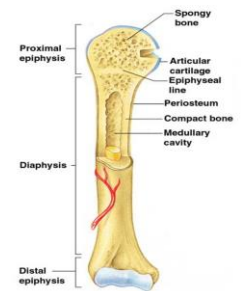
- Covers the external surface of the epiphyses



- Made of hyaline cartilage
- Decreases friction at joint surfaces

• Medullary cavity

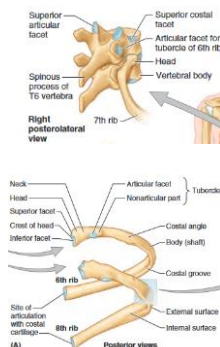
- Cavity of the shaft
- Contains red marrow (for blood cell formation) in infants



Cont....

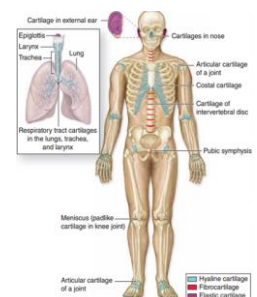
Bone Markings

- Surface features of bones
- Sites of attachments for muscles, tendons, and ligaments
- Passages for nerves and blood vessels
- Categories of bone markings
- Projections and processes: grow out from the bone surface.
- Depressions or cavities: indentations



Cartilage

- connective tissue composed of chondrocytes and extracellular matrix
- Unlike rigid bone, cartilage has the flexibility of firm plastic
- Three types:
 - ✓ Fibrous cartilage
- forms at the meniscus and the inter-vertebral disks



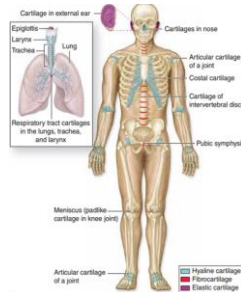
Cont....

✓ **Hyaline cartilage**

- covers **articular** bone surfaces (where one or more bones meet at a joint), connects the **ribs** and sternum, and appears in the **trachea**, **bronchi**, and **nasal septum**

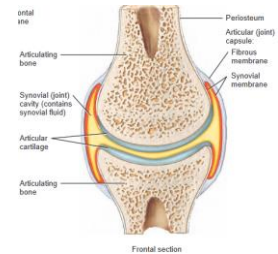
✓ **Elastic cartilage**

- located in the **auditory canal**, **external ear**, and **epiglottis**



Joints

- Articulations
- point of contact:
 - b/n bones
 - b/n cartilage and bones
 - b/n teeth and bones
- Classified:



Functionally

- **Synarthroses** = immovable joints
- **Amphiarthroses** = slightly moveable joints
- **Diarthroses** = freely moveable joints

Structurally

- **Fibrous joints** = Generally immovable
- **Cartilaginous joints**
 - Immovable or slightly moveable
- **Synovial joints** = Freely moveable



Cont....

Muscular system

- consists of all the muscles of the body
- Constitutes *~half of body's weight*
- Skeletal muscle about **40%** of body weight in men & **32%** in women
- consists of about 600 muscles



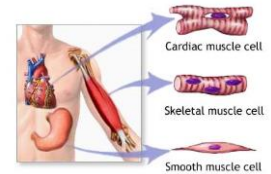
Cont....

- ✓ Movement
- ✓ Maintenance of posture
- ✓ Heat production (thermogenesis)
- ✓ Stabilization of joints



Types of muscle tissue

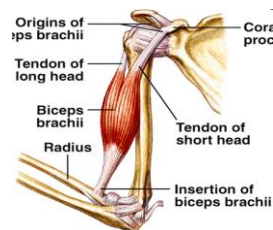
- Based on striations muscles could be:
Striated or Non-striated
- Based on location can be classified as:
Skeletal, Cardiac & Smooth muscles
- Based on function:
Voluntary VS Involuntary



Skeletal Muscles

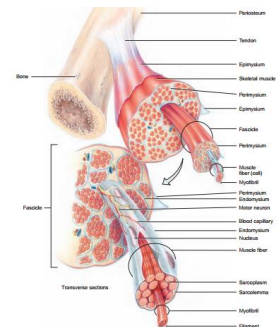
- composed of hundreds to thousands of cells, which are called **muscle fibers**

- proximal attachment = origin** of a muscle
- The more movable (that moves most) = **distal attachment = insertion** of a muscle



Connective Tissue Investment of Muscles

- Endomysium**: cover the individual muscle fibers
- Perimysium**: binds groups of fibers together into bundles called **fasciculi**
- Epimysium**: covers the entire muscle



Muscle Architecture

1. Parallel:

- Fascicles parallel to **longitudinal** axis of muscle; terminate at either end in flat.
- E.g. sartorius



2. Sphincter:

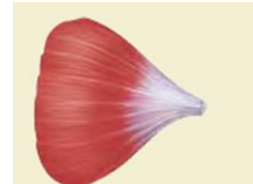
- surround a body opening or orifice.
- E.g. orbicularis oris



Cont....

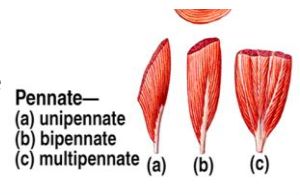
3. Convergent:

- fan-shaped muscles.
- E.g. deltoid



4. Pennate:

- feather shaped
- muscles have many fibers per unit area & are strong muscles (unipennate, bipennate & multipennate)

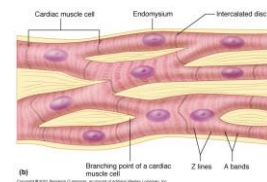


Nerves and Blood vessels

- In general, each skeletal muscle is supplied by **one** nerve, **one** artery & **one** or **more** veins
- Each **muscle fiber** in a skeletal muscle is contacted by **one nerve** ending, which signals the fiber to contract such a contact is called **neuromuscular junction**.
- A skeletal muscle fiber **cannot** contract unless it is stimulated by a nerve impulse

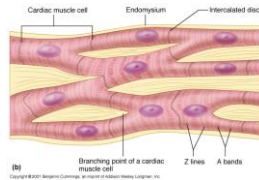
Cont....

- muscle tissue of the **heart wall**
- involuntary muscle
- Contraction pumps blood through the blood vessels
- **striated**
- muscle fibers are **branched** longitudinally



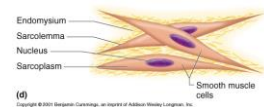
Cont....

- Ends of adjacent cells are connected through an intercellular junctional complex called the **intercalated disc**
- contain one or two centrally located nuclei



Smooth muscle

- Found in the walls of visceral organs such as the GIT, respiratory tubes, urinary bladder, uterus
- spindle shaped cell
- one centrally located nucleus
- non-striated
- actively divide and regenerate



Naming of muscles

1. according to their function

- Flexor E.g. Flexor carpi radialis
- Adductor E.g. Adductor longus

2. according to their attachment

- Scm
- Sternothyroid

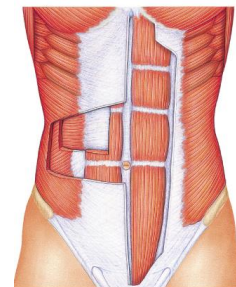


3. according to their shape

- Deltoid E.g. Deltoid
- Serratus E.g. Serratus anterior

4. according to orientation of fibers

- Rectus E.g. rectus abdominis
- Transverse E.g. transverse abdominis



Cont....

Cont....

5. according to their relative size

- Maximus E.g. gluteus maximus
- Longus E.g. flexor digitorum longus

6. according to the number of heads

- biceps brachii
- triceps brachii
- quadriceps

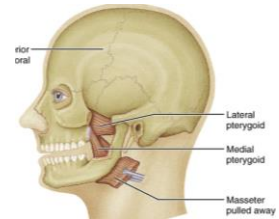


7. according to relative position

- Medial
- internal

8. Naming with a combination of criteria

- flexor carpi radialis longus
- extensor digiti minimi



Cont....

Functions of muscles

- Muscles serve specific functions in moving and positioning the body

I. prime mover (agonist)

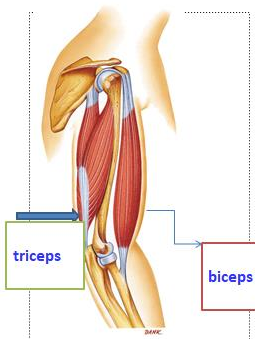
- is the main muscle responsible for producing a specific movement of the body

II. synergist

- complements the action of a prime mover

III. antagonist

- is a muscle that opposes the action of another muscle



Distribution of skeletal muscles

muscles of the axial skeleton

- responsible for facial expression, mastication, eye movement, tongue movement, respiration; and those of the abdominal wall and the pelvic outlet.

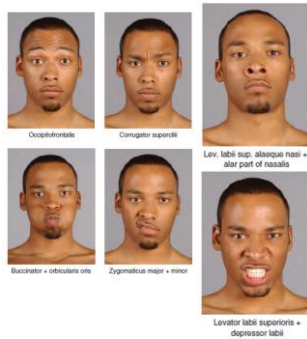
muscles of the appendicular skeleton

- Includes muscles of the pectoral girdle, arm, forearm, wrist, hand, pelvic girdle, thigh, leg and foot

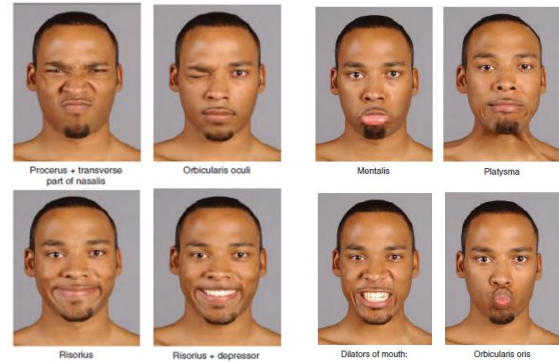
Muscles of the Axial Skeleton

I. Muscles of Facial Expression

- social communication.
- their origins are in the fascia or in the bones of the skull, with insertions into the **skin**
- **don't move** the facial skeleton.
- *receive their motor innervations from the facial nerve (CN VII)*

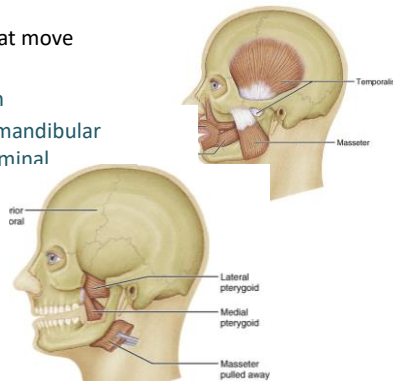


Cont....



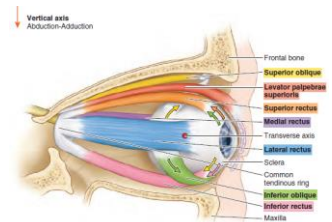
II. Muscles Of Mastication

- Are muscles that move the **mandible**
- assist in **speech**
- innervated by **mandibular branch of trigeminal nerve**.



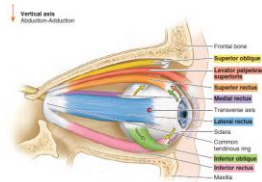
III. Ocular Muscles

- Two types of muscles are associated with the eyeball,
 - Extrinsic
 - intrinsic
- Extrinsic muscles originate **outside** the eyeball and are inserted on its **outer surface**
- They **move** the eyeballs in various directions.



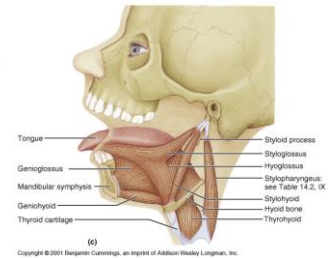
Cont....

- They **move** the eyeballs in various directions
- Intrinsic muscles originate and insert **entirely** within the eyeball
- They move structures within the eyeballs, such as the **iris** and the **lens**.



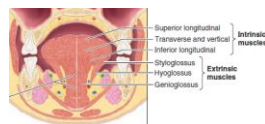
IV. Muscles of the Tongue

- the intrinsic and extrinsic muscles
- supplied by **hypoglossal nerve (CN XII)** except the **palatoglossus**,
- which is supplied by the **pharyngeal branch of vagus nerve (CN X)**



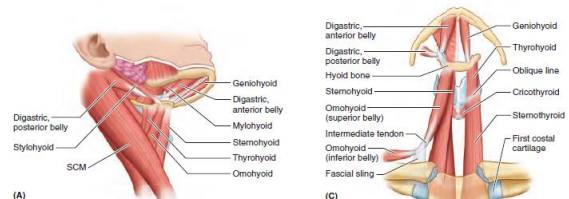
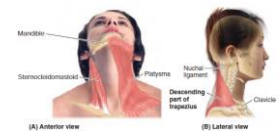
Cont....

- **four** intrinsic and extrinsic muscles **in each half** of the tongue are separated by **lingual septum**



V. Muscles of the Neck

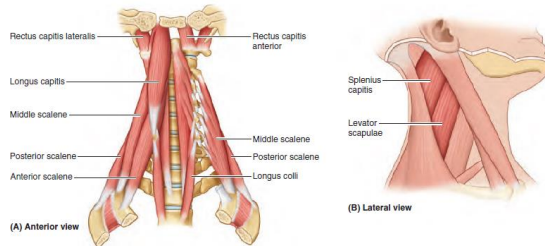
1. Superficial Neck Muscles
 2. Muscles of anterior cervical region
- ✓ Suprahyoid muscles
 - ✓ Infrahyoid muscles



Cont....

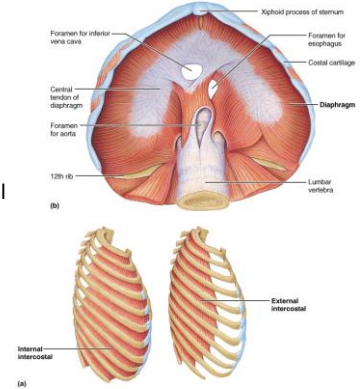
3. Prevertebral muscles

- Are the anterior and lateral vertebral muscles



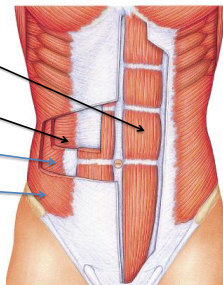
VI. Muscles of Respiration

- Pectoralis major
- Serratus anterior
- Serratus posterior
- External intercostal
- Internal intercostal
- Innermost intercostal
- Transversus thoracis
- Diaphragm



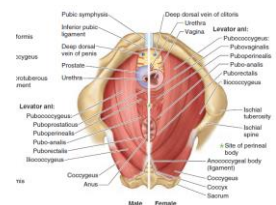
VII. Muscles of the Abdominal Wall

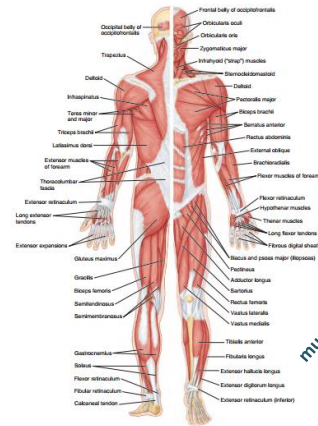
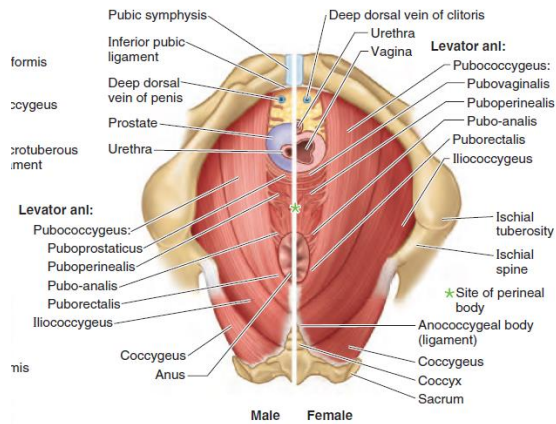
- Rectus abdominis
- Transverse abdominis
- Internal and external oblique abdominis
- pyramidalis



VIII. Muscles of the Pelvic Outlet

- Pelvic outlet supports the pelvic viscera
- Consists of **levator ani** & **coccygeus** muscles together called **pelvic diaphragm**
- **Perineal muscles** are positioned inferior to the pelvic diaphragm
- Innervated by **pudendal nerve**



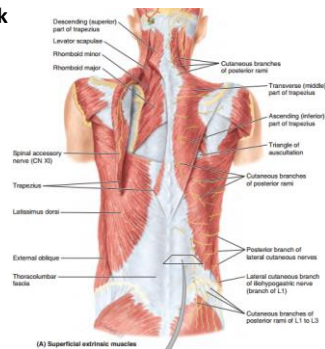


muscles of the appendicular skeleton

1. Back muscles

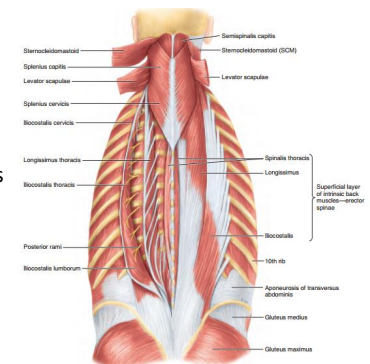
i. Superficial extrinsic back muscles

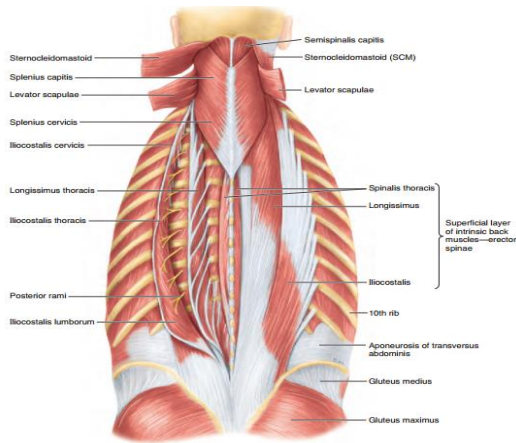
- are posterior axio-
appendicular
- Trapezius
- latissimus dorsi
- levator scapulae
- rhomboids



- ii. **intrinsic back muscles**

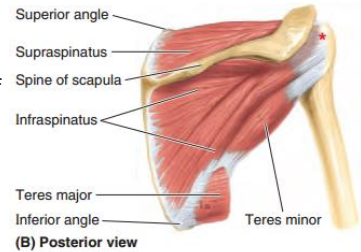
- splenius muscles
- erector spinae muscles
- transversospinalis muscle group





iii. Scapulohumeral (intrinsic shoulder) muscles:

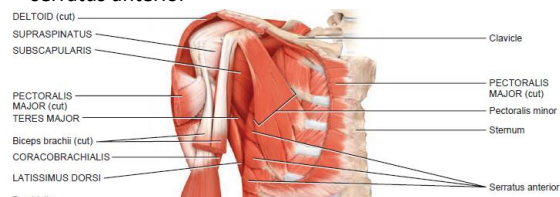
- Deltoid
- teres major
- four rotator cuff muscles
- Supraspinatus
- Infraspinatus
- teres minor,
- subscapularis



Cont....

iv. anterior axio - appendicular muscles

- move the pectoral girdle
- Pectoralis major
- pectoralis minor
- Subclavius
- serratus anterior



Cont....

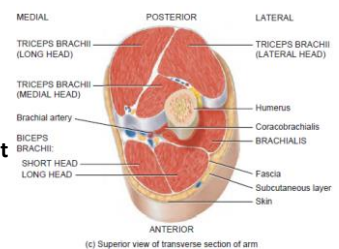
2. Arm muscles

i. Anterior compartment

- biceps brachii
- Brachialis
- Coracobrachialis

ii. posterior compartment

- triceps brachii



3. Muscles of Forearm

1. flexor/pronator muscles

A. superficial layer

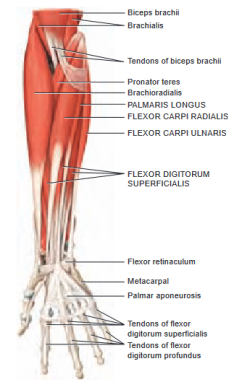
- pronator teres
- flexor carpi radialis
- palmaris longus
- flexor carpi ulnaris

B. intermediate layer

- flexor digitorum superficialis

C. deep layer

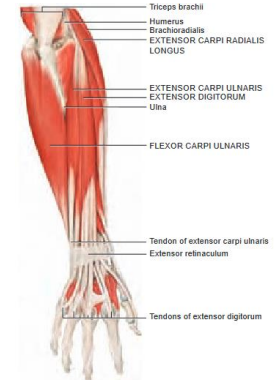
- flexor digitorum profundus
- flexor pollicis longus
- pronator quadratus



Cont....

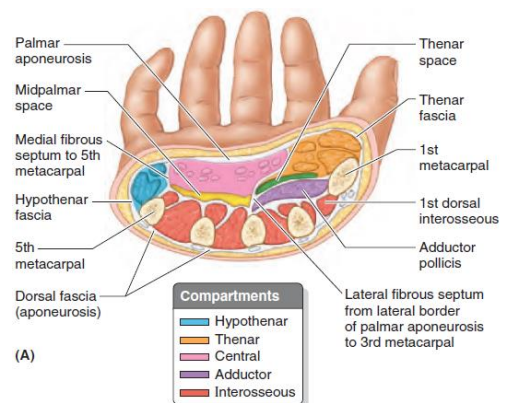
2. Extensor muscles

- extensor carpi radialis longus
- extensor carpi radialis brevis
- extensor carpi ulnaris



4. Muscles of Hand

- Thenar muscles in the **thenar compartment**:
- abductor pollicis brevis, flexor pollicis brevis, and opponens pollicis
- Adductor pollicis in the **adductor compartment**
- Hypothenar muscles in the **hypothenar compartment**:
- abductor digiti minimi, flexor digiti minimi brevis, and opponens digiti minimi
- Short muscles of the hand, the lumbricals, in the **central compartment** with the long flexor tendons.
- The interossei in separate **interosseous compartments** between the metacarpals



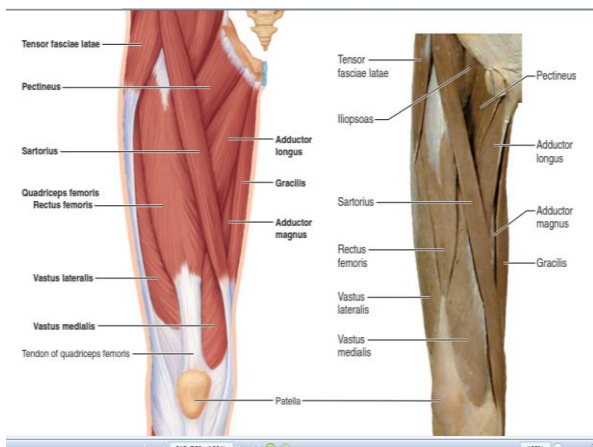
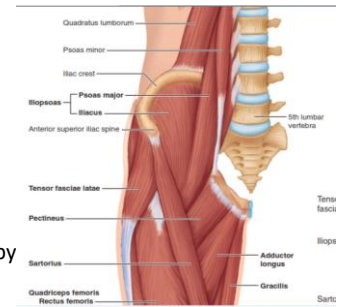
Muscles of the thigh

- Originated from the pelvic girdle & inserted to different sites on the femur
- Stabilize a highly movable hip joint
- Provide support for the body during bipedal stand and locomotion
- The muscles are divided in to
 - ✓ Anterior
 - ✓ posterior
 - ✓ and medial groups

Anterior Thigh Muscles

- flexors of the **hip** and
extensors of the **knee**

- include the :
 - ✓ Pectineus
 - ✓ Iliopsoas
 - ✓ Sartorius
 - ✓ quadriceps femoris

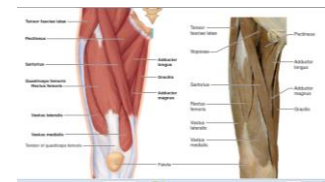


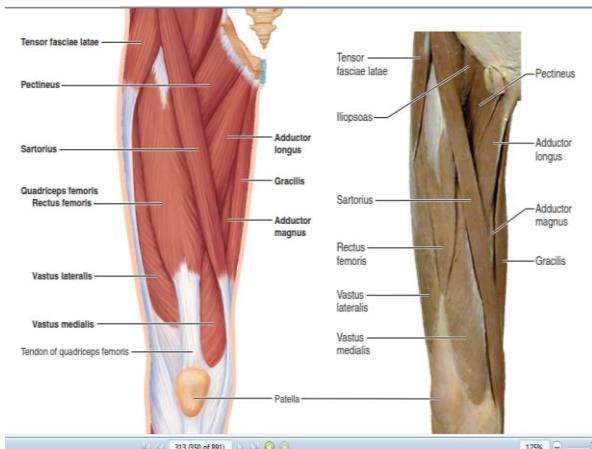
Medial thigh muscles

- Comprise the **adductor group**

- consisting of the:
 - ✓ adductor longus
 - ✓ adductor brevis
 - ✓ adductor magnus
 - ✓ Gracilis
 - ✓ Obturator externus

- All except the "hamstring part" of the adductor magnus and part of the pectineus are supplied by the obturator nerve



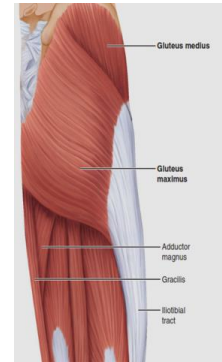


Muscles of the gluteal region

- are organized into two layers

I. superficial layer of muscles of the gluteal region

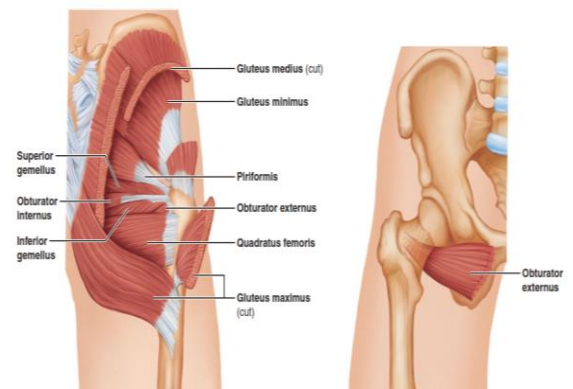
- consists of
- ✓ Gluteus maximus
- ✓ Gluteus medius
- ✓ Gluteus minimus
- ✓ Tensor fasciae latae



Cont....

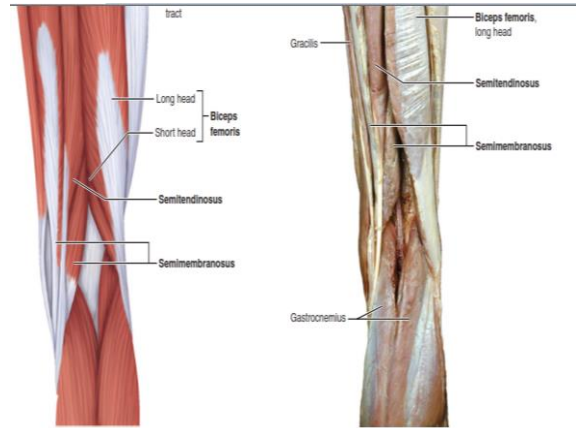
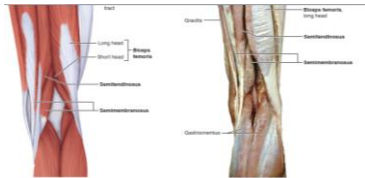
II. deep layer of muscles of the gluteal region

- consists of:
- ✓ Piriformis
- ✓ obturator internus
- ✓ Superior and inferior gemelli,
- ✓ quadratus femoris



Posterior thigh muscles

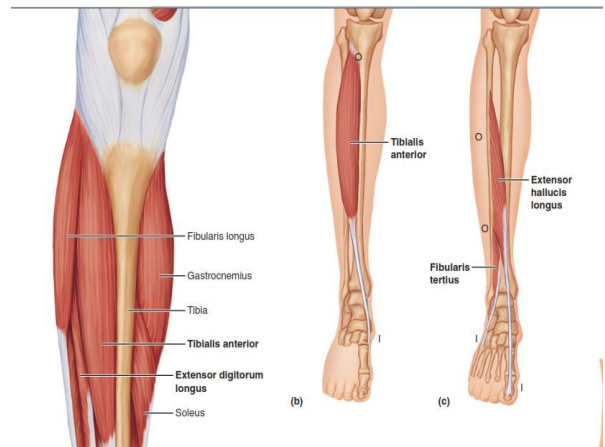
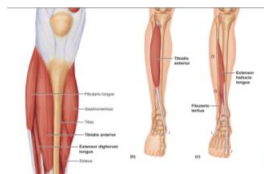
- Are extensors of hip and flexors of knee
- Include:
 - ✓ Semitendinosus
 - ✓ Semimembranosus
 - ✓ Biceps femoris
- Generally innervated by the tibial portion of the sciatic nerve



Muscles of the leg

Muscles of anterior compartment of leg

- Are the
 - ✓ *tibialis anterior*
 - ✓ *extensor digitorum longus*
 - ✓ *extensor hallucis longus*
 - ✓ *fibularis tertius*
- Innervated by deep fibular nerve



Cont....

Muscles in lateral compartment of leg

- contains the:
- ✓ *fibularis longus*
- ✓ *brevis muscles*
- Innervated by the superficial fibular nerve



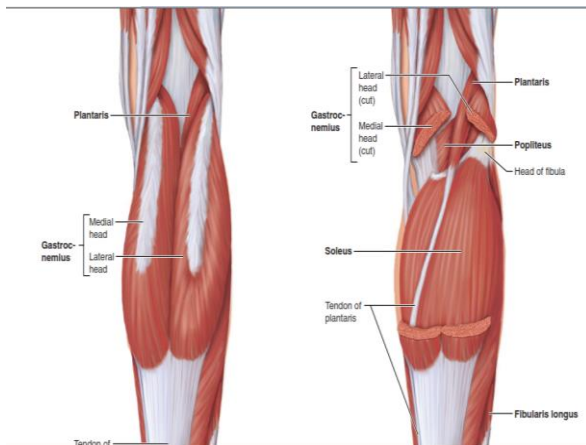
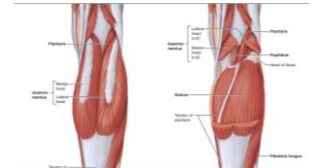
Cont....

Muscles in posterior compartment of the leg

- divided into superficial and deep subcompartments/ muscle groups
- Innervated by the tibial nerve

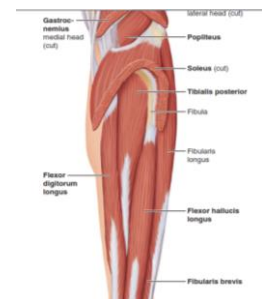
I. Superficial muscle group in posterior leg compartment

- includes the :
- ✓ *Gastrocnemius*
- ✓ *Soleus*
- ✓ *plantaris*

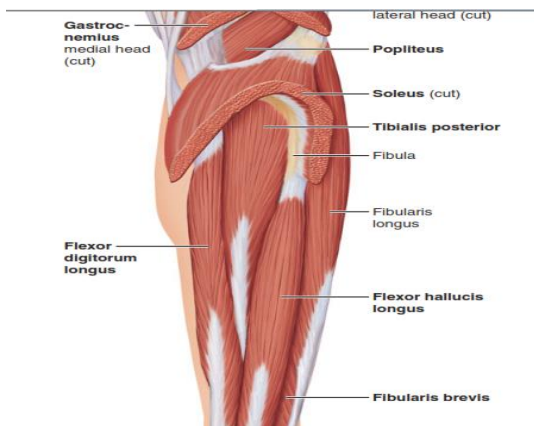


II. Deep muscle group in posterior compartment

- Include:
- ✓ *Popliteus*
- ✓ *flexor digitorum longus*
- ✓ *flexor hallucis longus*
- ✓ *tibialis posterior*



Cont....



Muscles of Foot

- Muscles on the dorsum of foot
- Include:
 - ✓ Extensor digitorum brevis
 - ✓ Extensor hallucis brevis
- Innervated by deep fibular nerve



Cont....

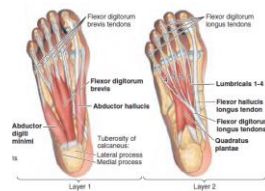
- muscles on the foot the sole

1st layer

- ✓ Abductor hallucis
- ✓ Flexor digitorum brevis
- ✓ Abductor digiti minimi

2nd layer

- ✓ Quadratus plantae
- ✓ Lumbricals



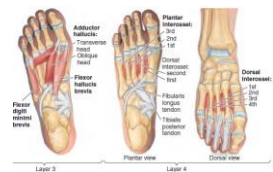
Cont....

3rd layer

- ✓ Flexor hallucis brevis
- ✓ Adductor hallucis
- ✓ Flexor digiti minimi brevis

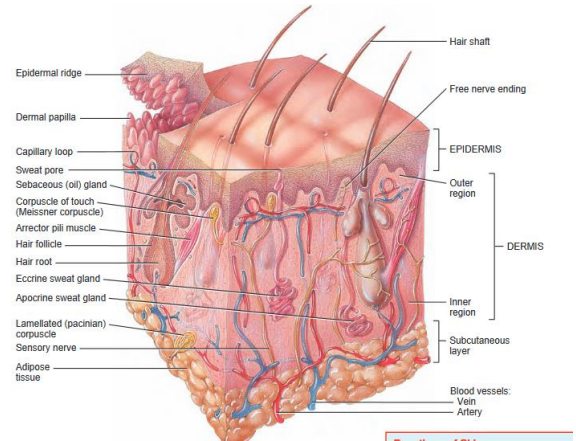
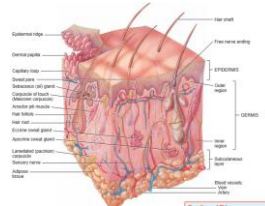
4th layer

- ✓ Plantar interossei
- ✓ Dorsal interossei



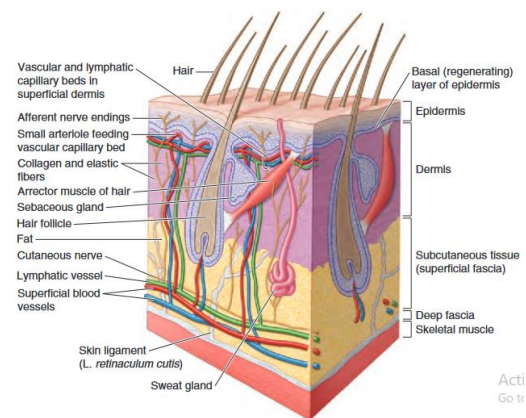
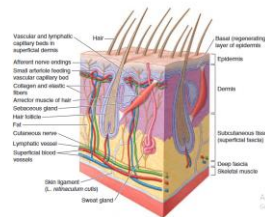
The integumentary system

- Include the skin and its derivatives
- sweat glands
- sebaceous glands
- Hairs
- nails



Functions of the integumentary system

- Protection
- temperature homeostasis
- Receives stimuli
- Excretion
- Synthesis of Vitamin D
- Reservoir for Blood
- nonverbal communication



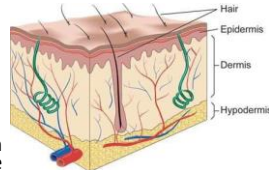
Skin

- is the largest single organ of the body
- account 15–20% of total body weight
- Consists of:

- I. Epidermis
- II. Dermis

➤ Hypodermis

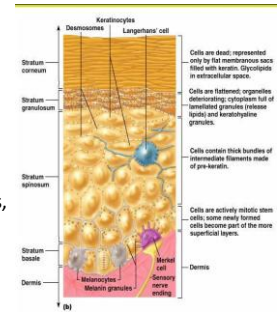
- beneath the dermis which attaches the dermis to the underlying structures
- not part of the skin but has close association with it



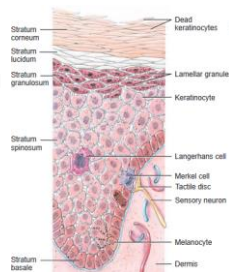
Cont....

I. Epidermis

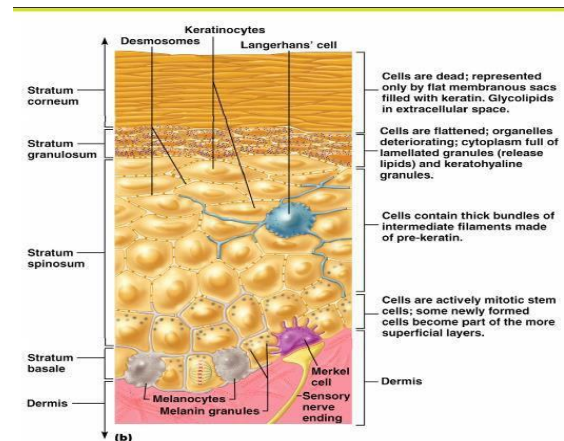
- Outermost superficial region
- Composed of epithelial cells
- Avascular
- Composed of **keratinized stratified squamous** epithelium, consisting of **4 distinct cell types** and **4 or 5 layers**
- Cell types include: keratinocytes, melanocytes, Merkel cells, and Langerhans' cells



- **Keratinocytes**: most cells in epidermis synthesize **keratin**
- **Melanocytes**: in the deepest layer synthesize **pigment melanin**
- **Merkel cells**: **touch receptors**, attached to nerve cell
- **dentic cells (Langerhans cells)**: are **macrophages** that stand guard against toxins, microbes and other pathogens that penetrate the skin
- if detected they alert immune system



Cont....



Layers of the Epidermis

Cont....

1. basal layer (stratum basale)

- Deepest layer firmly attached to the **Dermis**
- Consists of a single row of basophilic **columnar** or **cuboidal** cells
- Characterized by intense **mitotic** activity
- Responsible for constant production of **epidermal cells** in conjunction with the initial portion of the next layer

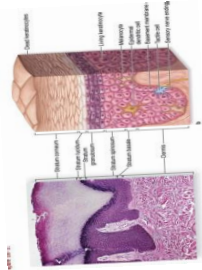
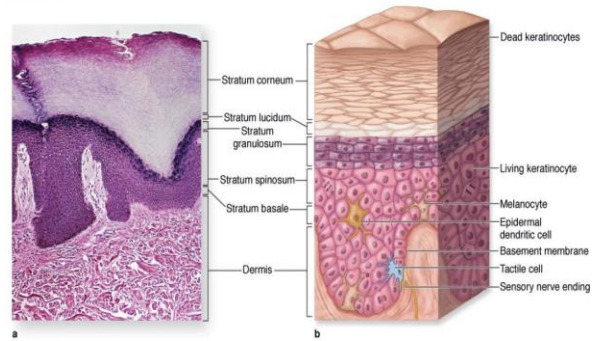


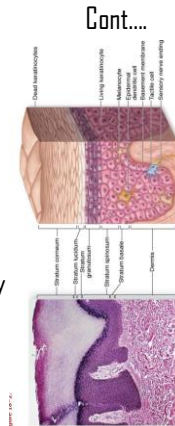
Figure 10-2.



- **Melanocytes** and **Merkel** cells are present

2. Spinous layer (stratum spinosum)

- **thickest** epidermal layer
- Mitosis is **less** frequent than in the stratum basale
- Just above the basal layer some cells may still divide and this, combined zone is called the stratum **germinativum**

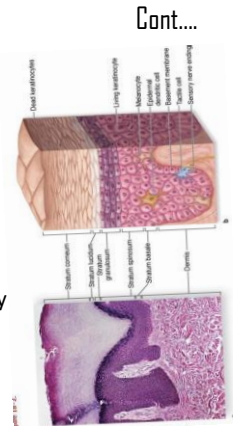


Cont....

- Melanin granules and **Langerhans'** cells are abundant in this layer

3. The granular layer (stratum granulosum)

- consists of **3–5** layers of flattened polygonal cells undergoing **terminal differentiation**
- as cells move up from s. basale they **die** & get flatter,
- and **Keratinization** begins in this layer



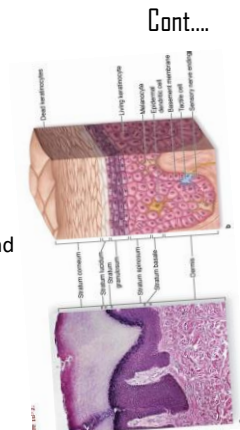
Cont....

4. Stratum lucidum

- only seen in **thick** skin
- translucent layer of extremely flattened **eosinophilic** cells

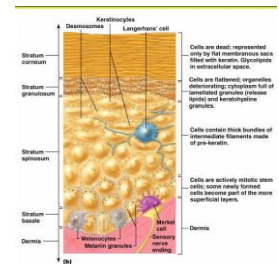
5. Stratum corneum

- consists of **15–20** layers of flattened, **nonnucleated** **keratinized** cells
- The cells are continuously **shed** and replaced by cells that move up from the lower layers
- **protects** the lower layers of the epidermis from water loss and chemical, physical and biological damage



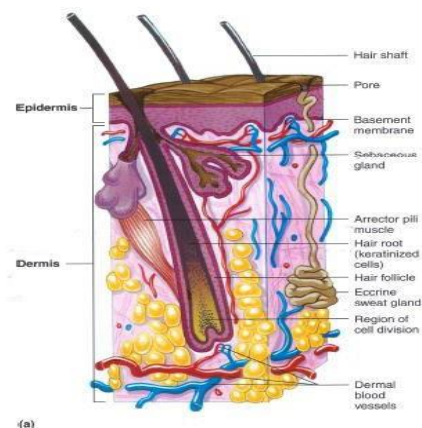
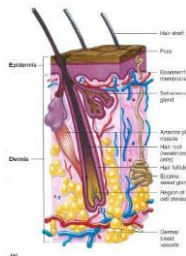
- **Thick** skin is found on the palm, fingertips, and soles of the feet.

- **Thin** skin is found on the rest of the body where the,
- - **stratum lucidum** is absent and the other 4 layers are thinner.



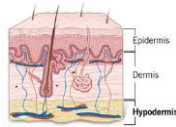
Dermis

- located below the basement membrane
- binds to the **hypodermis**
- has a rich supply of **blood** vessels, **nerve** fibers, **sense receptors** and **lymphatic** vessels
- Thickness varies according to the region of the body, and reaches its **maximum** of **4 mm** on the back
- **Hair** follicles, **sebaceous** glands, and **sweat** glands are found in the dermis, but they are produced by cells from the epidermis
- composed of the **papillary** and **reticular** layers



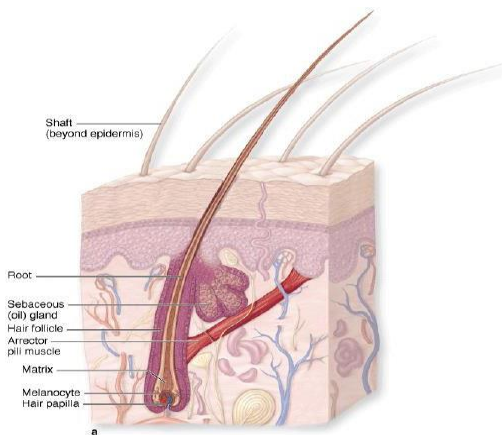
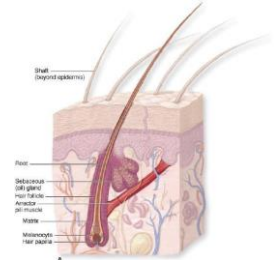
Hypodermis

- Deepest region / subcutaneous
- Also called the **superficial** fascia (superficial to the fascia of the skeletal muscles)
- Consists mostly of **connective** tissue
- Anchors the skin to the underlying muscle but loosely enough that the skin can slide freely
- contains **fat** cells that vary in number in different regions of the body and vary in size according to **nutritional** state



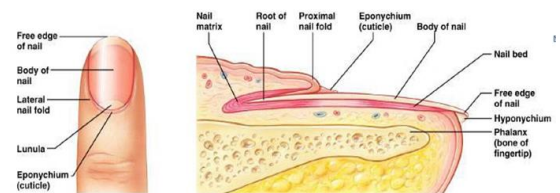
Hair

- Present on all skin surfaces except the **palms, soles, lips, nipples** and parts of **external** reproductive organs
- Extend from skin surface into the dermis and contain the hair root
- Hair is composed of **dead** epidermal cells
- **Arrector pili** muscle (bundle of smooth muscle cells) attaches to each hair follicle
- Hairs grow **discontinuously**, with periods of growth followed by periods of rest



Structure of nail

- Scale like modification of the epidermis on the distal, dorsal surface of the fingers and toes



Glands of the skin

❖ Sebaceous Glands

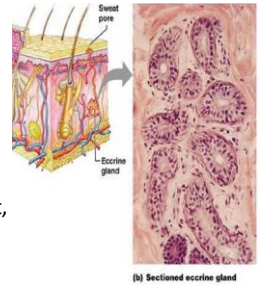
- oil glands
- located all over the body except on the palms of hands and soles of the feet
- usually associated with hair follicles, but some open through pores in the epidermis of the skin
- secrete an oily substance called sebum
- Sebum functions in moisturizing hair, softening and lubricating the skin,
- preventing water loss from evaporation, and in destroying bacteria



- Found over entire surface of body except on the nipple and parts of the external genitalia
- Two types:
 - eccrine
 - apocrine

A. Eccrine sweat glands (Merocrine)

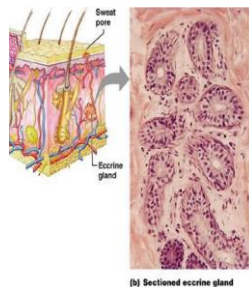
- Found in palms, soles of the feet, and forehead as well as over most of the body



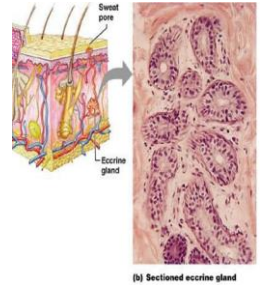
Cont....

Cont....

- Coiled, tubular gland
- Secretory part is coiled in dermis
- Duct opens as a funnel-shaped pore at the skin surface
- Secretes sweat,
 - a hypotonic filtrate of the blood that passes through the secretory cells of the sweat glands
 - released by exocytosis



- Regulate by the sympathetic autonomic nervous system
- Role is to cool the body (thermoregulation)
- Sweat is 99% H₂O and contains NaCl, vitamin C, antibodies, urea, uric acid, ammonia, pH4-6

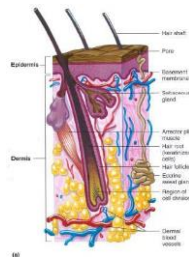


Cont....

Cont....

B. Apocrine sweat glands

- Found in **axillary** and **anogenital** areas
- Larger than eccrine glands
- Ducts empty into **hair** follicles
- Contains same components as true sweat as well as **proteins** and fatty **substances** making it **viscous**
- When its organic molecules are degraded by bacteria on the skin, it gives a musky smell (body odor)
- Begin functioning at **puberty**



Cont...

✓ Ceruminous glands

- modified **apocrine** glands located in the **external ear** canal below the sebaceous gland
- open through pores in the **epidermis** or into the **sebaceous** ducts
- The combined secretions of the sebaceous glands and the ceruminous glands is called **cerumen**